Q1

```
Start
    Function  solution(A, K)

    IF A is empty
        Return A
    End IF

    Repeat  K times
        last = last element of A
        Remove last element from A
        Insert last at the beginning of A
    End  Repeat

    Return A
    End Function
End
```

Q2

```
Start
    Function solution(X, Y, D)

    distance = Y - X

    jumps = distance DIV D

    If distance MOD D is not 0
        jumps = jumps + 1
    End if

    Return jumps

    End function
End
```

Q3

```
Start
    Function  solution (A)

    total = sum of all numbers in A
    left = 0
    minDiff = infinity

    For i from 0 to length of A - 2
        left = left + A[i]
        right = total - left

    diff = absolute value of (left - right)

        If diff < minDiff
            minDiff = diff
        End if
    End for

    Return minDiff

    End function
End
```

Q4

```
Start
    Function solution(A)
    N = length of A
    Sort A in ascending order
    For i from 0 to N-1
        If A[i] is not equal to i+1
            Return 0
        End if
    End for
    Return 1
    End function
End
```

Q5

```
Start
    Function solution(A)
    east = 0
    pairs = 0
    For each car in A
        If car == 0
            east = east + 1
        Else
            pairs = pairs + east
        end if
        If pairs > 1000000000
            Return -1
        end if
    End for
    Return pairs
    End function
End
```